

Add one-sided quota bounds to population-monotone ex ante proportional apportionment methods

ChatGPT (gpt-5.6-sol)*

Solution generated on 2026-08-24

Abstract

This paper gives a complete solution to the quota question for randomized apportionment methods that are ex ante proportional and whose deterministic realizations are universally population-monotone. For two states, an explicit randomized method satisfies both lower and upper quota at every input. For every number of states $n \geq 3$, lower quota is impossible, and upper quota is likewise impossible, even when the two quota requirements are considered separately. Thus no case of the stated problem remains unresolved.

1 The open problem

This paper addresses an open problem posed by Paul Gözl, Dominik Peters, Ariel D. Procaccia in *In This Apportionment Lottery, the House Always Wins* [1] (Economics and Computation (EC), 2022), Page 8 (PDF page 10), end of Section 3.2 (immediately after Theorem 2 and discussion). The website catalogue entry is problem 26 (W4285092914_p0) of the [Open Problems in Operations Research](#) collection.

Determine whether there exists a randomized apportionment method F that satisfies population monotonicity and ex ante proportionality and, in addition, satisfies at least one of the following one-sided quota properties:

- 1) *Lower quota: $F_i(p, h) \geq \lfloor q_i(p, h) \rfloor$ almost surely for all (p, h) and $i \in N$.*
- 2) *Upper quota: $F_i(p, h) \leq \lceil q_i(p, h) \rceil$ almost surely for all (p, h) and $i \in N$.*

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

*Generated by the `base_solver_ni_two_iterations` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve $\rightarrow$ A.8 verify $\rightarrow$ A.9 write-up; run `run_20260824_092502_a262897f, ec-2it-ultra`); two-iteration run, outcome from iteration 1. Independent verifier assessment: APPROVED with progress score 3/3 (full solution). Maintained by the AI in Mathematics & Operations Research project.

2 Model and complete result

Fix an integer $n \geq 2$ and write

$$[n] = \{1, \dots, n\}.$$

Populations are vectors

$$p = (p_1, \dots, p_n) \in \mathbb{N}^n,$$

where $\mathbb{N} = \{1, 2, \dots\}$, and the house size is an integer $h \in \mathbb{N}$. Let

$$P(p) = \sum_{i=1}^n p_i,$$

and define the exact quota of state i by

$$q_i(p, h) = h \frac{p_i}{P(p)}.$$

In particular,

$$\sum_{i=1}^n q_i(p, h) = h.$$

An allocation at (p, h) is a vector

$$a = (a_1, \dots, a_n) \in \mathbb{Z}_{\geq 0}^n \quad \text{such that} \quad \sum_{i=1}^n a_i = h.$$

We adopt the canonical solution-valued model of randomization: a randomized apportionment method is a probability distribution over deterministic rules. Equivalently, on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ there is a random rule

$$F(\cdot, \cdot; \omega)$$

such that, for every $\omega \in \Omega$, every $p \in \mathbb{N}^n$, and every $h \in \mathbb{N}$,

$$F(p, h; \omega) \in \mathbb{Z}_{\geq 0}^n, \quad \sum_{i=1}^n F_i(p, h; \omega) = h.$$

A single ω determines the rule simultaneously on the entire input domain; in particular, random choices are not resampled independently at different inputs.

The method is *ex ante proportional* if, for every p , h , and i ,

$$\mathbb{E}[F_i(p, h)] = q_i(p, h).$$

A deterministic rule f is *universally population-monotone* if, for every pair of inputs (p, h) and (p', h') and every pair of distinct states i, j , the inequalities

$$p'_i \geq p_i, \quad p'_j \leq p_j$$

preclude the simultaneous inequalities

$$f_i(p', h') < f_i(p, h), \quad f_j(p', h') > f_j(p, h).$$

Thus a state whose population has not decreased cannot lose a seat simultaneously with a gain by a state whose population has not increased. The randomized method is universally population-monotone if almost every deterministic realization has this property.

The method satisfies *lower quota* if, for every p , h , and i ,

$$F_i(p, h) \geq \lfloor q_i(p, h) \rfloor \quad \text{almost surely.}$$

It satisfies *upper quota* if, for every p , h , and i ,

$$F_i(p, h) \leq \lceil q_i(p, h) \rceil \quad \text{almost surely.}$$

Because $\mathbb{N}^n \times \mathbb{N}$ is countable, all almost-sure assertions concerning individual inputs, coordinates, or pairs of inputs can be imposed simultaneously after discarding a single null set: the relevant family of events is countable, and a countable intersection of probability-one events has probability one. Consequently, in the impossibility proofs below we may fix a common probability-one set on which universal population monotonicity and every quota inequality used in the argument hold simultaneously. All event inclusions are therefore understood up to a null set.

The complete answer is the following.

Theorem 1 (Complete classification). *Under the canonical solution-valued randomization model:*

1. *If $n = 2$, there exists an ex ante proportional, universally population-monotone randomized method satisfying both lower and upper quota at every input.*
2. *If $n \geq 3$, no ex ante proportional, universally population-monotone randomized method can satisfy lower quota.*
3. *If $n \geq 3$, no ex ante proportional, universally population-monotone randomized method can satisfy upper quota.*

Thus the problem is solved completely.

3 A consequence of universal population monotonicity

Let e_i denote the i -th standard basis vector of $\mathbb{Z}^n$.

Lemma 1 (House monotonicity). *Every universally population-monotone deterministic rule f is componentwise nondecreasing in the house size. More precisely, for every population vector p and every $h \in \mathbb{N}$, there is a unique state $K = K(p, h)$ such that*

$$f(p, h + 1) - f(p, h) = e_K.$$

Proof. Fix p and h . Suppose, toward a contradiction, that some state i loses a seat:

$$f_i(p, h + 1) < f_i(p, h).$$

Since the total allocation increases from h to $h + 1$,

$$\sum_{k=1}^n (f_k(p, h + 1) - f_k(p, h)) = 1.$$

The negative change in coordinate i therefore forces the existence of a state $j \neq i$ satisfying

$$f_j(p, h+1) > f_j(p, h).$$

The populations at the two inputs are identical, so

$$p_i \geq p_i, \quad p_j \leq p_j,$$

while state i loses a seat and state j gains one. This contradicts universal population monotonicity.

It follows that every coordinate of

$$f(p, h+1) - f(p, h)$$

is a nonnegative integer. The coordinates sum to one, so the difference is e_K for a unique $K \in [n]$. $\square$

4 The construction for two states

Assume $n = 2$. Let U be uniformly distributed on $[0, 1]$, sampled once to determine the entire deterministic rule. For $0 \leq u < 1$, define

$$F_1(p, h; u) = \lfloor q_1(p, h) + u \rfloor, \tag{1}$$

$$F_2(p, h; u) = h - F_1(p, h; u) = \lceil q_2(p, h) - u \rceil. \tag{2}$$

At the null endpoint $u = 1$, define

$$F_1(p, h; 1) = \lceil q_1(p, h) \rceil,$$

$$F_2(p, h; 1) = \lfloor q_2(p, h) \rfloor.$$

The equality in (2) follows from $q_1(p, h) + q_2(p, h) = h$ and the identity

$$m - \lfloor x \rfloor = \lceil m - x \rceil,$$

valid for every integer m and every real x .

Proposition 1. *For every $p \in \mathbb{N}^2$, every $h \in \mathbb{N}$, and every $u \in [0, 1]$, the vector $F(p, h; u)$ is an allocation of h seats and satisfies both lower and upper quota:*

$$\lfloor q_i(p, h) \rfloor \leq F_i(p, h; u) \leq \lceil q_i(p, h) \rceil \quad (i = 1, 2).$$

Proof. The definitions give

$$F_1(p, h; u) + F_2(p, h; u) = h$$

both for $u < 1$ and for $u = 1$.

Consider first $0 \leq u < 1$, and write

$$q_1(p, h) = m + r, \quad m \in \mathbb{Z}_{\geq 0}, \quad 0 \leq r < 1.$$

Then

$$m \leq m + r + u < m + 2,$$

so

$$F_1(p, h; u) = \lfloor m + r + u \rfloor \in \{m, m + 1\}.$$

If $r = 0$, then $m + r + u < m + 1$, and hence $F_1 = m$. If $r > 0$, then

$$\lfloor q_1 \rfloor = m, \quad \lceil q_1 \rceil = m + 1.$$

In either case,

$$\lfloor q_1 \rfloor \leq F_1 \leq \lceil q_1 \rceil.$$

The same conclusion for state 2 follows directly from

$$F_2 = \lceil q_2 - u \rceil.$$

Indeed, if $q_2 = m + r$ with $0 \leq r < 1$, then

$$m - 1 < m + r - u \leq m + 1,$$

and a case distinction between $r = 0$ and $r > 0$ gives

$$\lceil q_2 - u \rceil \in \{\lfloor q_2 \rfloor, \lceil q_2 \rceil\}.$$

At $u = 1$, the definitions give

$$F_1 = \lceil q_1 \rceil, \quad F_2 = \lfloor q_2 \rfloor,$$

so both quota inequalities hold. Moreover,

$$\lceil q_1 \rceil + \lfloor q_2 \rfloor = \lceil q_1 \rceil + \lfloor h - q_1 \rfloor = h.$$

Thus the endpoint also yields a valid allocation satisfying both quota bounds. $\square$

Proposition 2. *The two-state method is ex ante proportional.*

Proof. Fix a real number $x = m + r$, where $m \in \mathbb{Z}$ and $0 \leq r < 1$. Apart from the probability-zero endpoint $U = 1$,

$$\lfloor x + U \rfloor = \begin{cases} m, & 0 \leq U < 1 - r, \\ m + 1, & 1 - r \leq U < 1, \end{cases}$$

where the second event is empty when $r = 0$. Therefore

$$\mathbb{P}(\lfloor x + U \rfloor = m + 1) = r$$

and

$$\mathbb{E}[\lfloor x + U \rfloor] = m(1 - r) + (m + 1)r = m + r = x.$$

The special convention at $U = 1$ does not affect the expectation. Applying this identity to $x = q_1(p, h)$ yields

$$\mathbb{E}[F_1(p, h)] = q_1(p, h).$$

Since $F_1 + F_2 = h$ identically,

$$\mathbb{E}[F_2(p, h)] = h - \mathbb{E}[F_1(p, h)] = h - q_1(p, h) = q_2(p, h). \quad \square$$

Proposition 3. *Every deterministic realization of the two-state method is universally population-monotone.*

Proof. Fix $u \in [0, 1]$. If $u < 1$, then

$$F_1(p, h; u) = \lfloor q_1(p, h) + u \rfloor$$

is a nondecreasing function of $q_1(p, h)$, while

$$F_2(p, h; u) = \lceil q_2(p, h) - u \rceil$$

is a nondecreasing function of $q_2(p, h)$. At $u = 1$, the functions

$$F_1(p, h; 1) = \lceil q_1(p, h) \rceil, \quad F_2(p, h; 1) = \lfloor q_2(p, h) \rfloor$$

are likewise nondecreasing functions of the respective individual quotas.

Suppose that two inputs (p, h) and (p', h') and distinct states $i, j \in \{1, 2\}$ satisfy

$$p'_i \geq p_i, \quad p'_j \leq p_j,$$

but that a forbidden reversal occurs:

$$F_i(p', h'; u) < F_i(p, h; u), \quad F_j(p', h'; u) > F_j(p, h; u).$$

Because the allocation of each state is a nondecreasing function of that state's exact quota, these strict allocation inequalities imply

$$q_i(p', h') < q_i(p, h), \quad q_j(p', h') > q_j(p, h).$$

All quotas are positive, so

$$\frac{q_i(p', h')}{q_j(p', h')} < \frac{q_i(p, h)}{q_j(p, h)}.$$

For each input, the common house-size and total-population factors cancel in the ratio of two quotas. Hence

$$\frac{p'_i}{p'_j} = \frac{q_i(p', h')}{q_j(p', h')} < \frac{q_i(p, h)}{q_j(p, h)} = \frac{p_i}{p_j}.$$

On the other hand, positivity of the populations together with $p'_i \geq p_i$ and $p'_j \leq p_j$ gives

$$\frac{p'_i}{p'_j} \geq \frac{p_i}{p_j},$$

a contradiction. Thus no forbidden reversal is possible. □

Propositions 1, 2, and 3 together prove the first part of Theorem 1.

5 Impossibility of lower quota for $n \geq 3$

Theorem 2. *For every $n \geq 3$, there is no ex ante proportional, universally population-monotone randomized apportionment method satisfying lower quota.*

Proof. Fix $n \geq 3$ and suppose, toward a contradiction, that such a method exists. Consider the population vectors

$$p = (1, n, \underbrace{1, \dots, 1}_{n-2 \text{ entries}})$$

and

$$p' = (2, n, \underbrace{1, \dots, 1}_{n-2 \text{ entries}}).$$

Their total populations are

$$P(p) = 1 + n + (n - 2) = 2n - 1$$

and

$$P(p') = 2 + n + (n - 2) = 2n.$$

Only the population of state 1 changes; all other populations are unchanged.

Let

$$A = \{\omega : F(p, 1; \omega) = e_3\}$$

and

$$B = \{\omega : F(p', 1; \omega) = e_3\}.$$

We first prove the almost-sure inclusion

$$A \subseteq B.$$

Fix a realization in the common probability-one set on which universal population monotonicity, house monotonicity, and all required lower-quota inequalities hold, and suppose that this realization belongs to A . Thus state 3 receives the unique seat at $(p, 1)$.

Let k be the state that receives the seat at $(p', 1)$. Suppose first that

$$k \notin \{1, 3\}.$$

At $(p, 1)$, state 3 has one seat and state k has none, whereas at $(p', 1)$, state 3 has no seat and state k has one. Both states have unchanged populations:

$$p'_3 = p_3, \quad p'_k = p_k.$$

Thus state 3, whose population has not decreased, loses a seat while state k , whose population has not increased, gains a seat. This contradicts universal population monotonicity.

The only possible winner other than state 3 is therefore state 1. Assume that

$$F(p', 1) = e_1.$$

At the input $(p', 2)$, the exact quota of state 2 is

$$q_2(p', 2) = 2 \frac{p'_2}{P(p')} = 2 \frac{n}{2n} = 1.$$

Lower quota consequently requires

$$F_2(p', 2) \geq \lfloor 1 \rfloor = 1.$$

By Lemma 1, the allocation is componentwise nondecreasing when the house grows from one seat to two. Since state 1 already holds one seat at $(p', 1)$,

$$F_1(p', 2) \geq 1.$$

There are only two seats in total, so necessarily

$$F(p', 2) = e_1 + e_2.$$

Now compare $(p, 1)$ with $(p', 2)$. At the first input, state 3 has one seat and state 2 has none; at the second input, state 3 has no seat and state 2 has one. Their populations are unchanged:

$$p'_3 = p_3, \quad p'_2 = p_2.$$

This is again a forbidden population-monotonicity reversal. Hence state 1 cannot win at $(p', 1)$, and the winner must be state 3, which proves $A \subseteq B$ up to a null set.

At house size one, every allocation is a standard basis vector. Therefore

$$\mathbf{1}_A = F_3(p, 1), \quad \mathbf{1}_B = F_3(p', 1).$$

Ex ante proportionality gives

$$\mathbb{P}(A) = \mathbb{E}[F_3(p, 1)] = q_3(p, 1) = \frac{1}{2n-1}$$

and

$$\mathbb{P}(B) = \mathbb{E}[F_3(p', 1)] = q_3(p', 1) = \frac{1}{2n}.$$

The almost-sure inclusion $A \subseteq B$ requires

$$\mathbb{P}(A) \leq \mathbb{P}(B),$$

but

$$\frac{1}{2n-1} > \frac{1}{2n}.$$

This contradiction proves the theorem. The calculation and every case above remain valid when $n = 3$. $\square$

6 Impossibility of upper quota for $n \geq 3$

Theorem 3. *For every $n \geq 3$, there is no ex ante proportional, universally population-monotone randomized apportionment method satisfying upper quota.*

Proof. Fix $n \geq 3$ and suppose, toward a contradiction, that such a method exists. Define

$$p = (1, 1, n, \underbrace{1, \dots, 1}_{n-3 \text{ entries}})$$

and

$$p' = (1, 2, n, \underbrace{1, \dots, 1}_{n-3 \text{ entries}}).$$

For $n = 3$, the displayed tails are empty. Set

$$S = P(p) = 1 + 1 + n + (n - 3) = 2n - 1.$$

Then

$$P(p') = 1 + 2 + n + (n - 3) = 2n = S + 1.$$

Only the population of state 2 changes.

At the input (p, S) , every exact quota is integral:

$$q_i(p, S) = S \frac{p_i}{P(p)} = p_i.$$

Upper quota gives

$$F_i(p, S) \leq p_i \quad (i \in [n]).$$

The coordinates on both sides have the same total:

$$\sum_{i=1}^n F_i(p, S) = S = \sum_{i=1}^n p_i.$$

Consequently, every coordinate inequality must be an equality, and hence

$$F(p, S) = p \quad \text{almost surely.}$$

Likewise,

$$q_i(p', S + 1) = p'_i,$$

and upper quota together with the total-seat constraint forces

$$F(p', S + 1) = p' \quad \text{almost surely.}$$

Work on the common probability-one set on which these identities, upper quota, universal population monotonicity, and Lemma 1 all hold. By house monotonicity, there are unique random states $K, K' \in [n]$ such that

$$F(p, S) - F(p, S - 1) = e_K$$

and

$$F(p', S + 1) - F(p', S) = e_{K'}.$$

Equivalently,

$$F(p, S - 1) = p - e_K$$

and

$$F(p', S) = p' - e_{K'}.$$

We claim that

$$\{K = 1\} \subseteq \{K' = 1\}$$

up to a null set.

Assume $K = 1$. Then

$$F(p, S - 1) = p - e_1 = (0, 1, n, 1, \dots, 1).$$

Suppose first that $K' = k$ for some $k \geq 3$. Then

$$F(p', S) = p' - e_k.$$

In passing from the input $(p, S - 1)$ to (p', S) , state k loses one seat:

$$F_k(p', S) = p'_k - 1 = p_k - 1 < p_k = F_k(p, S - 1),$$

while state 1 gains one seat:

$$F_1(p', S) = 1 > 0 = F_1(p, S - 1).$$

Both populations are unchanged:

$$p'_k = p_k, \quad p'_1 = p_1.$$

This contradicts universal population monotonicity. Hence K' cannot be any state $k \geq 3$.

It remains to exclude $K' = 2$. In this case,

$$F(p', S) = p' - e_2 = (1, 1, n, 1, \dots, 1) = p.$$

By house monotonicity, for some unique state L ,

$$F(p', S) - F(p', S - 1) = e_L.$$

Thus

$$F(p', S - 1) = p' - e_2 - e_L.$$

At this input, the exact quota of state 3 is

$$q_3(p', S - 1) = (S - 1) \frac{p'_3}{P(p')} = (2n - 2) \frac{n}{2n} = n - 1.$$

Upper quota therefore requires

$$F_3(p', S - 1) \leq \lceil n - 1 \rceil = n - 1.$$

Since

$$F_3(p', S) = n$$

and precisely one seat is removed in passing from house size S to house size $S - 1$, the removed seat must belong to state 3. Hence $L = 3$ and

$$F(p', S - 1) = p' - e_2 - e_3 = (1, 1, n - 1, 1, \dots, 1).$$

On the other hand,

$$F(p, S - 1) = p - e_1 = (0, 1, n, 1, \dots, 1).$$

Thus, between the inputs $(p, S - 1)$ and $(p', S - 1)$, state 3 loses one seat while state 1 gains one. Both populations are unchanged:

$$p'_3 = p_3, \quad p'_1 = p_1.$$

This again contradicts universal population monotonicity. Therefore $K' \neq 2$.

The only remaining possibility is $K' = 1$, which proves

$$\{K = 1\} \subseteq \{K' = 1\}$$

almost surely.

Because each relevant house-size increment is a unit vector,

$$\mathbf{1}_{\{K=1\}} = F_1(p, S) - F_1(p, S - 1) \quad \text{almost surely.}$$

Ex ante proportionality and $F_1(p, S) = p_1 = 1$ give

$$\begin{aligned} \mathbb{P}(K = 1) &= \mathbb{E}[F_1(p, S) - F_1(p, S - 1)] \\ &= q_1(p, S) - q_1(p, S - 1) \\ &= 1 - \frac{S - 1}{S} \\ &= \frac{1}{S}. \end{aligned}$$

Similarly,

$$\mathbf{1}_{\{K'=1\}} = F_1(p', S + 1) - F_1(p', S) \quad \text{almost surely,}$$

and hence

$$\begin{aligned} \mathbb{P}(K' = 1) &= \mathbb{E}[F_1(p', S + 1) - F_1(p', S)] \\ &= q_1(p', S + 1) - q_1(p', S) \\ &= 1 - \frac{S}{S + 1} \\ &= \frac{1}{S + 1}. \end{aligned}$$

The almost-sure inclusion requires

$$\mathbb{P}(K = 1) \leq \mathbb{P}(K' = 1),$$

whereas

$$\frac{1}{S} > \frac{1}{S + 1}.$$

This contradiction proves the theorem. The argument remains unchanged for the boundary case $n = 3$. $\square$

7 Conclusion

The two-state construction satisfies ex ante proportionality, universal population monotonicity, lower quota, and upper quota simultaneously. For every $n \geq 3$, Theorem 2 rules out lower quota, and Theorem 3 independently rules out upper quota under the same proportionality and population-monotonicity hypotheses. These results cover every allowed number of states $n \geq 2$ and therefore constitute a complete solution, with no remaining gap.

References

- [1] Paul Gözl, Dominik Peters, Ariel D. Procaccia. *In This Apportionment Lottery, the House Always Wins*. Economics and Computation (EC), 2022. <https://doi.org/10.1145/3490486.3538299>